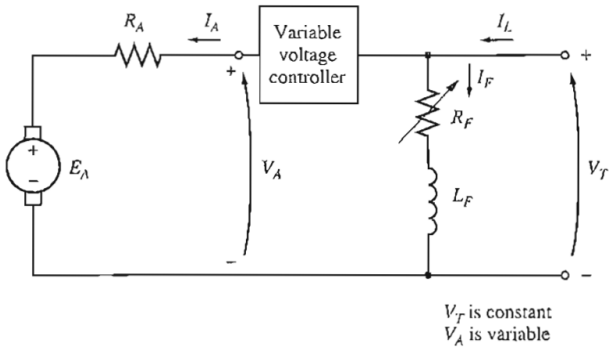
EE3042 Energy Conversion Final Exam, November 22th, 2022, Time Allowed: 75 minutes
Closed Notes and Books, Use only calculator, Not mobile phone. All necessary formulas are provided on the last page.

1. A linear machine has the following characteristics:

$$B = 0.8\,T\ into\ page, R = 0.25\ \Omega,\ \ l = 0.6\ \text{m},\ V_B = 100\ V$$

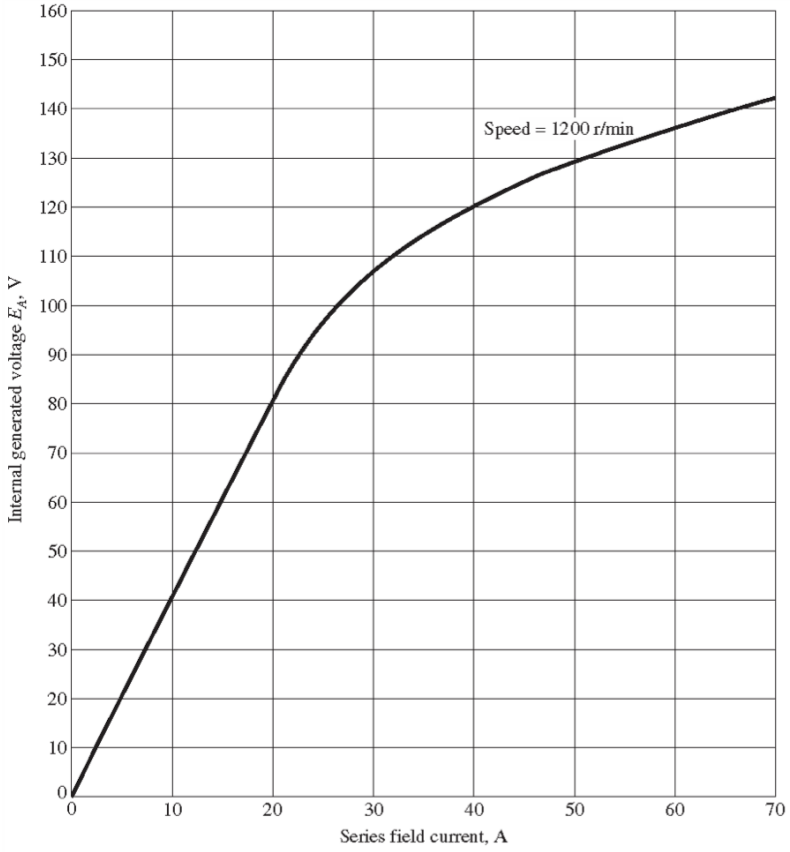
Explain the motor and generator actions by drawing a simple linear electrical machine. Write all the necessary equations describing the machine and sketch the plots of velocity, back emf voltage, current passing through the bar and force acting the bar, writing initial and steady-state values on the plots.

2. Consider the following circuit, which provides the speed control by both changing the armature voltage and field current. Explain these two techniques by taking into account the base speed (n_{base}), torque, power, field resistance, armature voltage and current. Draw the required graphical plots.



V_T is constant
V_A is variable

3. A 7.5-hp, 120-V series DC motor has an armature resistance of 0.12 Ω and a series field resistance of 0.07 Ω. At full load, the input current is 56 A, and the rated speed is 1075 rpm. The motor magnetization curve is the following. The core losses are 220 W, and mechanical losses are 230 W at full load. Suppose the mechanical losses vary as the speed of the motor by the fractional power of 3, i.e., $(speed)^3$. What are the speed and efficiency of the motor if it is operating at a 40-A armature current?



4. The magnetization curve for a separately excited DC generator is given below. The generator is rated at 6 kW, 122 V, 50 A, and 1800 r/min. The following data are known about the machine:

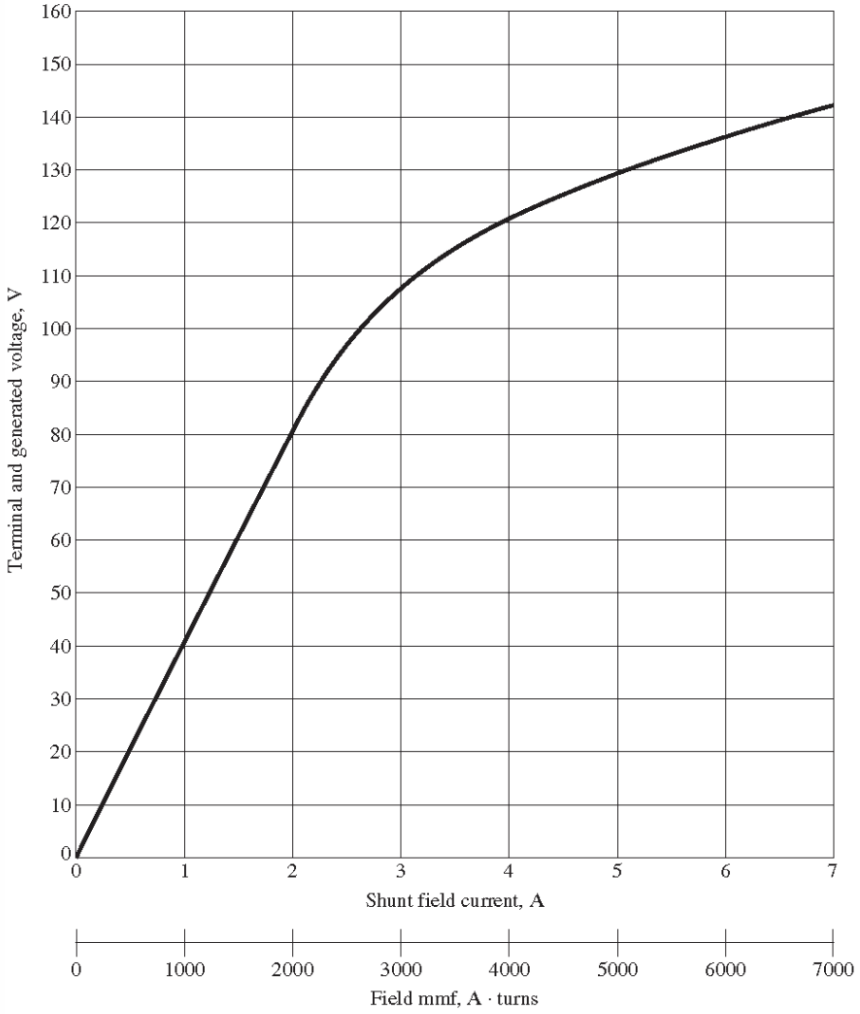
$$R_A = 0.17\ \Omega\,, \quad R_{adj} = 0\ to\ 40\ \Omega$$

$$V_F = 122\ V \quad R_F = 19.5\ \Omega, \qquad N_f = 1020\ turns\ per\ pole$$

Answer the following questions about this generator. assuming no armature reaction.

a) Draw the equivalent circuit. If this generator is operating at no load, what is the range of voltage adjustments that can be achieved by changing R_{adj}.

b) lf the field rheostat is allowed to vary from 0 to 30 Ω and the generator's speed is allowed to vary from 1500 to 2000 r/min, what are the maximum and minimum no-load voltages in the generator?

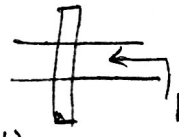


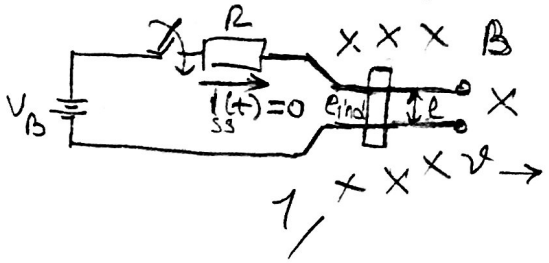
Output power	$P_{out} = P_{conv} - P_{mech} - P_{core}$
Power Converted to mechanical	$P_{conv} = E_A I_A$
Motor shaft power	$Pout = w\,\tau$
Generated voltage (bemf)	$E_A = K\emptyset w$
Core Reluctance	$\mathcal{R} = \dfrac{l}{\mu_o \mu_r A}$
Permeability in free space (μ_o)	$4\pi * 10^{-7}$

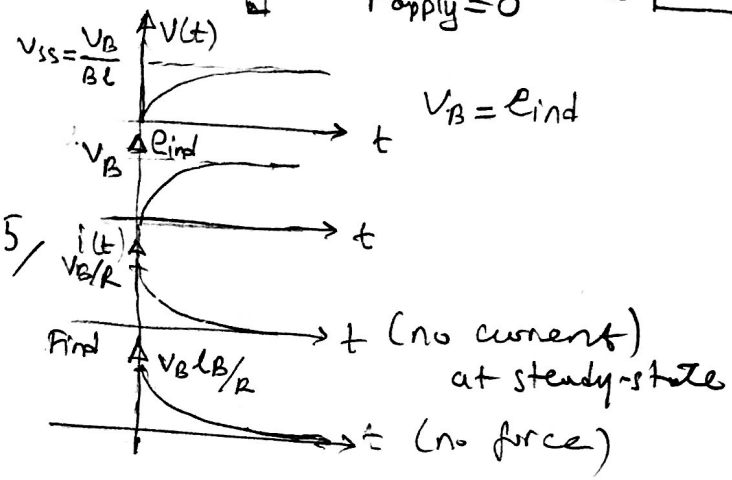
1. No load, motor, and generator action conditions for a linear DC m/c

i) No load condition

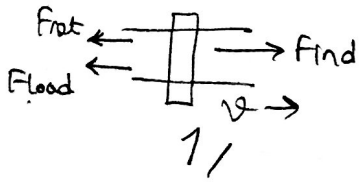
$F_{net} = 0$ no force applied & the bar at steady state







ii) Motor action



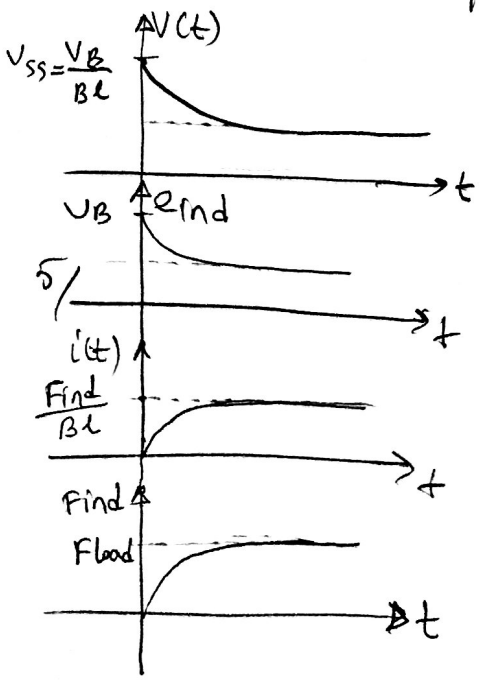
$P_{conv} = e_{ind} i = F_{ind} v$

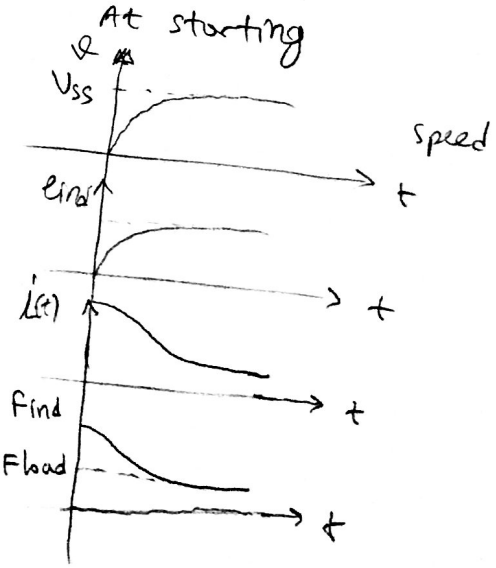
$e_{ind} = Bvl$, $F_{ind} = Bil$

$F_{net} = F_{load} - F_{ind} = 0$ at steady-state

Load condition: $\uparrow i = \frac{V_B - e_{ind}\downarrow}{R}$, $F_{ind}\uparrow$, $e_{ind} < V_B$

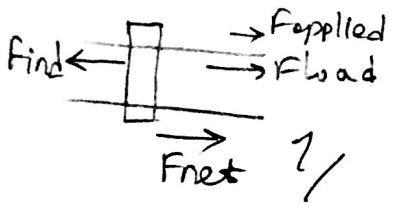
The induced force $F_{ind} = Bil$ increases until $F_{ind} = F_{load}$ at a lower speed.





Starting from no load to understand the motor action.

(iii) Generator action



Fload (Fapply)

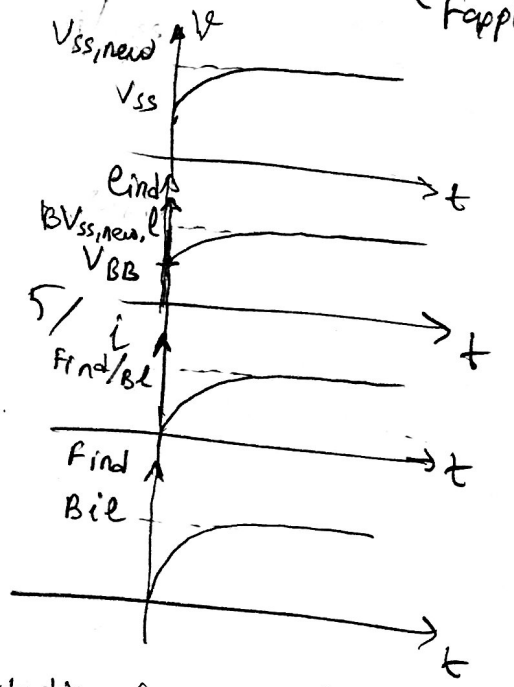
$e_{ind} > V_B$

. Find increases until Find = Fload

. The bar moves to the right at a higher ~~constant~~ speed wrt. steady-state.

e_{ind} increases ($e_{ind} = B v \ell$) and $i = \frac{e_{ind} - V_B}{R}$ also increases.

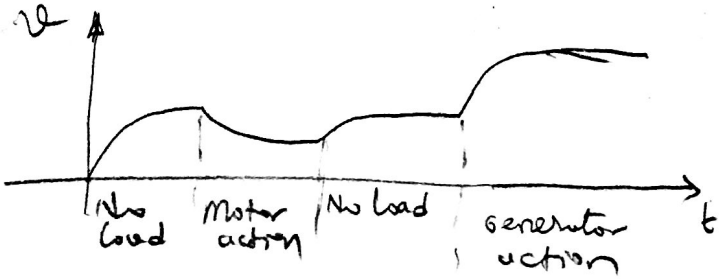
$\vec{F}_{net} = (\vec{F}_{load} + \vec{F}_{applied}) - \vec{F}_{ind} = 0$ at steady-state

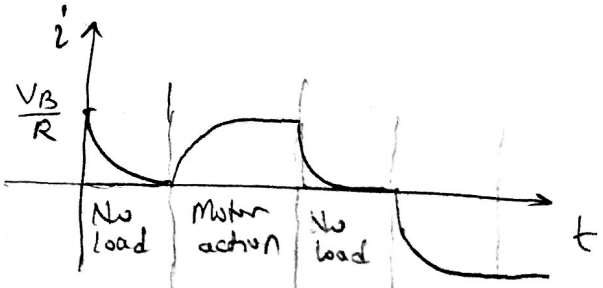


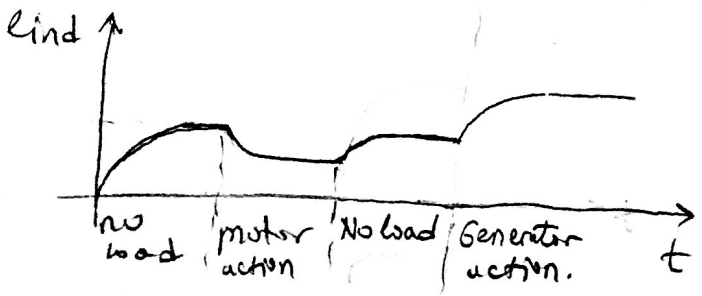
$P_{conv} = e_{ind} \cdot i = F_{ind} v_{ss,new}$

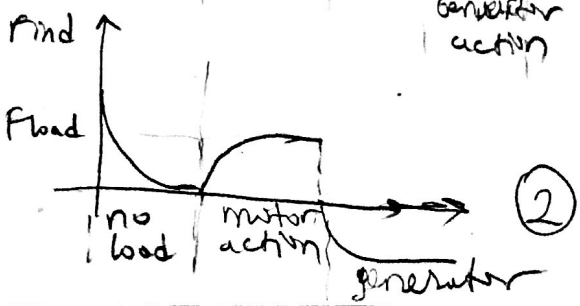
} the current in direction

starting from no load to understand the generator action.





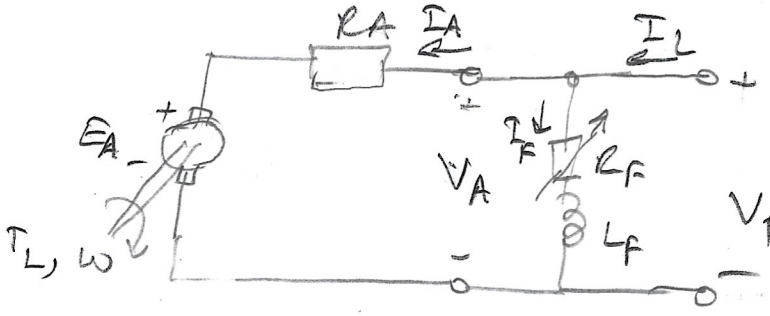




(2)

2) (25p)

2.) a) Speed Control by changing the armature voltage



Keeping V_T constant and V_A variable using a voltage controller

i) A change in V_A causes to change I_A

6/

$$I_A = \frac{V_A - E_A}{R_A} \quad (1)$$

$$\tau_{ind} = K\phi I_A \uparrow\downarrow \quad (2)$$

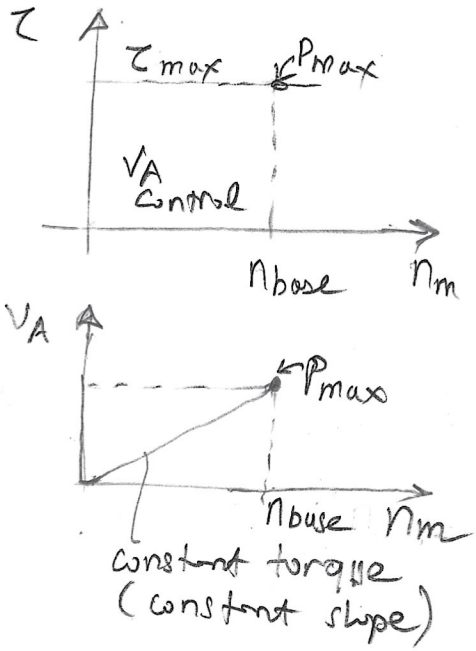
As soon as $\tau_{ind} > \tau_{load}$ or $\tau_{ind} < \tau_{load}$ ω_m will change (3)

ω_m causes to change: $E_A \uparrow\downarrow = K\phi \omega_m$ (4)

This time, E_A changes I_A : $I_A = \frac{V_T - E_A \uparrow\downarrow}{R_A}$ (5)

$\uparrow\downarrow I_A$ finally changes τ_{ind} until $\tau_{ind} = \tau_{load}$ at a different speed.

ii) 7/



Since V_T is kept constant, I_F will be constant. (the flux is now constant)

$\tau_{max} = K\phi I_{A,max}$

Loading the motor so that I_A is at maximum.

$P = E_A I_{A,max}$ E_A can be change between 0 and $E_{A,max}$ by changing V_A.

$P_{max} = \tau_{max} \omega_{base}$

2 b) Speed Control by changing the field current.

i) Changing R_F causes to $I_F (= V_T / R_F \uparrow\downarrow)$ change I_F (1)

6/ I_F changes the flux, ϕ (2)

ϕ changes $E_A (= K\phi \uparrow\downarrow \omega_m)$ (3)

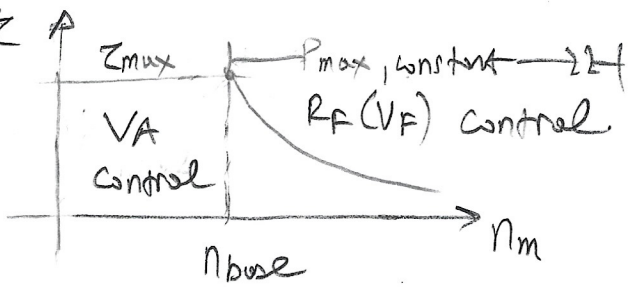
E_A changes $I_A (= \dfrac{V_T - E_A}{R_A})$ (4)

A change in I_A causes to change $\tau_{ind} (K\phi I_A)$. (5)
Where I_A is dominant over the change in flux.

So, τ_{ind} changes ω_m and ω_m causes to change E_A and E_A changes I_A until $\tau_{ind} = \tau_{load}$ at different speed (6)

ii)

6/



$P_{max} = E_{A,max} I_{A,max}$ (max or rated)

further increasing the speed beyond n_{base} τ will reduce as shown in the figure (left) since the power is maximum. $\left(\tau = \dfrac{P_{max}}{\omega_m}\right)$

3) Series DC MOTOR (25p)

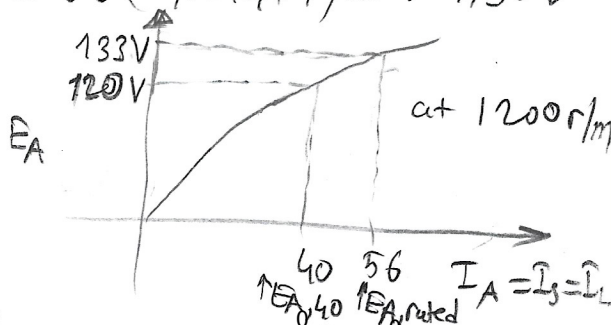
a) η% and n_m at $I_A = 40A$, $\begin{cases} n_{m,rated} = 1075 \text{ r/m} \\ I_{A,rated} = 56A \end{cases}$

2/ $P_{IN} = V_T I_L = (120)(40) = 4800W$

2/ $E_{A\,@40A} = V_T - I_A (R_A + R_S) = 120 - 40(0,12 + 0,07) = 112,4\,V$

2/ $E_{A,rated} = V_T - I_{A\,rated} (R_A + R_S) = 120 - 56(0,12 + 0,07) = 109,36\,V$

3/ $\dfrac{E_{A,rated}\ (56A)}{E_{A0,rated}\ (56A)} = \dfrac{n_{m,rated}}{n_{m0}}$



3/ $\dfrac{E_{A,40A}}{E_{A0,40A}} = \dfrac{n_{m,40A}}{n_{m0}}$ $\Rightarrow$ $\dfrac{E_{A,40A}}{E_{A,rated}} = \left(\dfrac{n_{m,40A}}{n_{m0}} E_{A0,40A}\right)\left(\dfrac{n_{m0}}{n_{m,rated}} \dfrac{1}{E_{A0,rated}}\right)$

4/ $n_{m,40A} = \left(\dfrac{E_{A,40A}}{\quad}\right)\left(\dfrac{E_{A0,rated}}{\quad}\right) n_{m,rated} = 1224,6 \text{ r/}$

3-continued) 2/ $P_{conv} = E_A I_A = (112,4)(40A) = 4496\,W$

$P_{out} = P_{conv} - P_{losses} = P_{conv} - P_{core} - P_{mech}$.

3/ $P_{mech\,(40A)} = (n_{m,40A})^3 \cdot \frac{P_{mech,rated}}{(n_{m,rated})^3} = \left(\frac{1224,6}{1075}\right)^3 (230\,W)$

$= 340\,W$

2/ $P_{out} = 4496 - 220 - 340 = 3936\,W = 3,936\,kW$

2/ $\eta\% = \frac{P_{out}}{P_{in}} \times 100 = \frac{3,936\,kW}{4,8\,kW} \times 100 = 82\,\%$

4) Separately excited DC gen.

(25p) $P_{out\,rated} = 6kW$, 122V, 50A, $n_{prime} = 1800\,r/mn$ with $I_{f\,rated} = 5A$

$R_A = 0,17\Omega$, $R_{adj} = 0$ to 40Ω, $R_f = 19,5\Omega$

$N_f = 1020$ turns/pole, No armature reaction

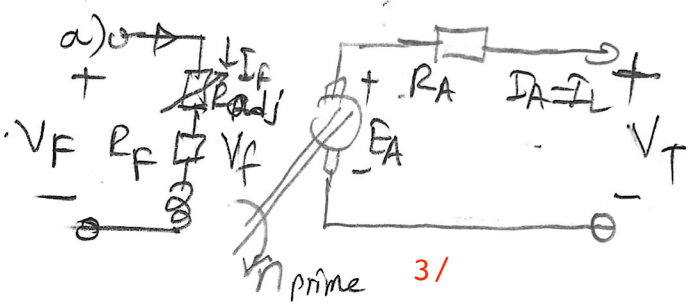
a) Equivalent circuit, $R_{adj} \rightarrow [0\ 40\Omega] \Rightarrow V_T = [V_0\ V_{40}]$

b) If R_f varied $\rightarrow [0\ 30\Omega]$

$n_{prime} \rightarrow [1500 - 2000\,r/min]$

$V_{T\,nl}$?

a)



3/ $I_{f\,0\Omega} = \frac{V_F}{R_{adj\,[0]} + R_f} = \frac{122V}{19,5} \cong 6,26\,A$

3/ $I_{f,40\Omega} = \frac{122V}{40 + 19,5} \cong 2\,A$

6/ $V_T = [V_{0\Omega}\ V_{40\Omega}] = [137V\ 80V]$

From the MC at no load with 1800 r/m.

b) $R_f \rightarrow [0\ 30\Omega]$

4. continued)

2/ $I_{f,max} = \frac{V_F}{R_f + R_{adj}} = \frac{122V}{19.5 + 0\Omega} \rightarrow$ ~~6.1~~ 6.26A $\rightarrow n_{prime\ 0\Omega} = 2000 r/m$

2/ $I_{f,min} = \frac{122}{19.5 + 30} = 2.46A \rightarrow n_{prime\ 30\Omega} = 1500 r/m.$

3/ $\frac{E_{A,0\Omega}}{E_{A0}} = \frac{n_{0\Omega}}{n_0} \Rightarrow E_{A,0\Omega} = \frac{2000 r/m}{1800 r/m}(137) = 152.2V = V_{T,nL}$

3/ $\frac{E_{A,30\Omega}}{E_{A0}} = \frac{n_{1,30\Omega}}{n_0} \Rightarrow E_{A,30\Omega} = \frac{1500 r/m}{1800 r/m}(94) \approx$ ~~78~~ 78.3 $V = V_{T,nL}$

The speeds of 2000 r/m and 1500 r/m are actual speeds.